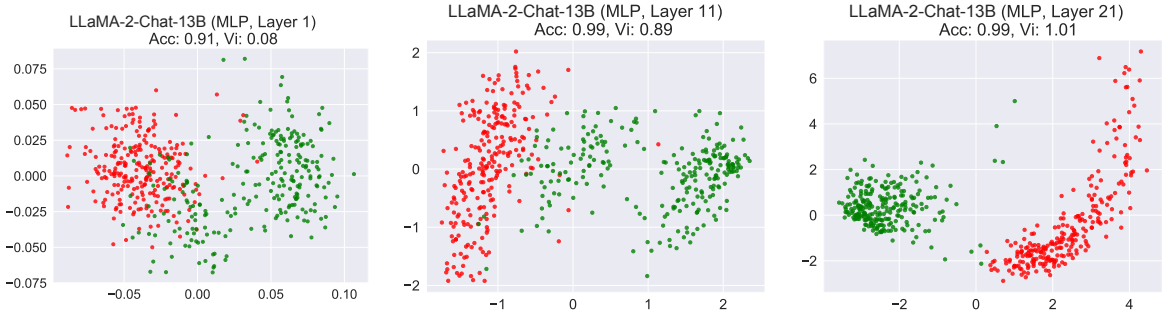


Figure 2: The principal component analysis (PCA) visualization of how context knowledge is processed across different layers in LLaMA 2 Chat 13B, where instances denoted green and red signify factual and counterfactual evidence, respectively. $\mathcal{V}$-usable information (Vi) is found to be more effective in distinguishing between dataset difficulty than accuracy (Acc).

where $H_{\mathcal{V}}(Y)$ and $H_{\mathcal{V}}(Y|R_w^{(i)})$ denote the predictive $\mathcal{V}$-entropy and the conditional $\mathcal{V}$-entropy, which can be estimated through the following equations:

$$H_{\mathcal{V}}(Y) = \inf_{f \in \mathcal{V}} \mathbb{E}[-\log_2 f[\varnothing](Y)], \tag{2}$$

$$H_{\mathcal{V}}(Y|R_w^{(i)}) = \inf_{f \in \mathcal{V}} \mathbb{E}[-\log_2 f[R_w^{(i)}](Y)], \tag{3}$$

where $\varnothing$ denotes an input devoid of any information, we employ zero-input vectors as an alternative representation. To minimize the influence of additional parameters, we constrain M_{probe} to linear classifier families when computing M_{probe}-usable information. Furthermore, we add 0.01 to the input before taking the logarithm to prevent highly anomalous values. Comparative results with test accuracy in Fig. 2 show that this validation metric exhibits a higher level of discriminative capability.

5. Experiments

5.1. Experimental Setup

Datasets We conduct our experiments based on the ConflictQA-popQA-gpt4 dataset (Xie et al., 2023). It constitutes a complementary extension of the entity-centric question-answering (QA) dataset popQA (Mallen et al., 2022a), which encompasses not only commonsense question-answer pairs and their associated popularity from Wikipedia but also incorporates parametric memories and counter-memories generated by GPT-4. We select 50 instances with both high popularity and consistency between parametric memories and ground truth. Subsequently, we generate 100 pieces of factual and counterfactual evidence for each fact based on parametric memory and counter-memory. For newly acquired knowledge which is discussed in Sec 5.4, we manually design a piece of knowledge that does not exist in parametric memory: *The password of the president's laptop is {password}*, and request `gpt-3.5-turbo-0613` to provide various evidence for different passwords. We randomly generate 4 distinct 10-character passwords and place them within the *{password}* placeholder. For each password, we generate 100 supporting pieces of evidence. For all settings, the temperature is set to 1.0 to generate diversity outputs.

Models We choose the open-access LLaMA 2 (Touvron et al., 2023) with 7, 13, and 70 billion parameters as the LLMs to be explained. During the probing phase, we adopt a linear classifier as our probing model to reduce extraneous interference. We use a batch size of 4, learning rate of 0.0001, Adam optimizer (Kingma and Ba, 2015), and 15 training epochs for all probing tasks. The ratio between training and test sets is 8:2. Unless otherwise stated, we conduct probing tasks on the Transformer layer output of each layer.

5.2. Preliminary Experiments: Can LLMs Retain Context Knowledge?

As a foundational step for subsequent experiments, it is imperative to ascertain whether the provided evidence has the capability to change answers generated by the LLM. We conduct an assessment of question-answering accuracy after providing different sets of evidence (Tab.1).

As can be seen in the table, the LLMs consistently provide correct answers when presented with factual evidence that aligns with the ground truth. Conversely, the accuracy drops approaching zero when the LLMs receive counterfactual evidence. These results indicate that the coherent and diverse evidence generated by ChatGPT can effectively alter the LLM's cognition of existing knowledge or facilitate the encoding of newly acquired knowledge.

Model Name	Conflicting				Newly Acquired	
	Factual ↑		Counterfactual ↓		Factual ↑	
	base	chat	base	chat	base	chat
LLaMA 2 7B	82.31	79.84	31.76	4.80	85.75	98.25
LLaMA 2 13B	83.82	80.60	17.27	6.08	91.50	98.25
LLaMA 2 70B	75.91	84.20	12.23	14.51	89.25	98.25

Table 1: Question-answering accuracy after providing different sets of evidence. In conflict scenarios, the average accuracy (%) is calculated after providing both factual and counterfactual evidence for various commonsense knowledge. In newly acquired scenarios, the 4-class accuracy is calculated based on the provided *{password}*.

However, we find that the base models maintain high accuracy even after providing counterfactual evidence, implying their low capability to encode knowledge through prompting. Therefore, the dataset we have constructed can more reliably induce the utilization of context knowledge in chat models, thereby enabling the exploration of the roles played by various layers within them. In subsequent experiments, we will conduct experiments mainly on chat models.

5.3. How Do LLMs Encode Conflicting Knowledge?

5.3.1. Case Study

Utilizing the probing method elucidated in Sec 4, we are able to obtain the layer-wise $\mathcal{V}$-information of each token in the LLM's processing of individual fact. We select the question *What is Mike Flanagan's occupation?* as a case study. We present the content of the factual and counterfactual evidence generated by ChatGPT, with the intermediary portions omitted (see Tab. 2). It can be seen that the generated content provides substantial support for the corresponding knowledge labels, even in cases of fictional content. Consequently, these texts serve as effective means to assess the capability of LLMs in encoding context knowledge. We generate 100 distinct instances of content for each knowledge label.

Then we investigate the output of the Transformer layers, MLP layers, and Attention layers separately on LLaMA 13B. We plot heatmaps depicting the layer-wise $\mathcal{V}$-information of LLaMA 13B while processing tokens in the given question (Fig. 3). It is evident that each component of LLaMA 13B exhibits significantly higher $\mathcal{V}$-information at upper layers, implying that context knowledge is encoded within their representations. For Transformer layer output and MLP layer output, most tokens maintain high and stable $\mathcal{V}$-information after 30 layers.

However, the results for Attention layer output appear relatively chaotic. This aligns with the observations made by Vig and Belinkov (2019). Multiple heads of self-attention may focus on distinct local or global information, thereby facilitating the transfer of information between token representations. This not only leads to the encoding of knowledge-related information in other tokens such as *What* and *'*, but also results in the propagation of irrelevant information within lower layers, causing a slight disruption in probing results.

Another intriguing discovery is that LLMs encode context knowledge to varying degrees within different tokens. For knowledge-related entity tokens such as *Mike* and *occupation*, LLMs achieve high $\mathcal{V}$-information at lower layers. This is not due to parametric memorization in LLMs, as simple parametric memorization would result in consistent behaviors when handling different external evidence, leading to lower $\mathcal{V}$-information. Therefore, the LLMs prioritize encoding context knowledge into knowledge-related entity tokens. In contrast, tokens that are not directly related to the knowledge such as *What* and *'*, encode relevant information later through the attention mechanism.

Validation of the broader applicability of these findings will be conducted in Sec 5.3.2 and 5.3.3.

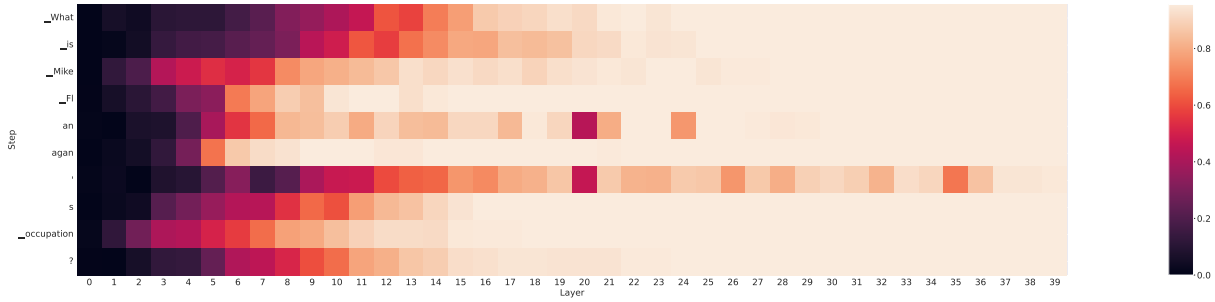
5.3.2. Average $\mathcal{V}$-information

We then conduct the probing tasks on all 50 facts and select the last token in the question to compute the average $\mathcal{V}$-information (Fig. 4). We observe that the $\mathcal{V}$-information gradually increases, especially in the early layers. This aligns with our conjecture that the LLM encodes more context knowledge within upper layers, enabling them to differentiate between facts and counterfactuals more effectively. Since the correct responses for facts and counterfactuals differ, the layers of LLMs tend to preserve differences accumulated from previous layers and capture more context knowledge.

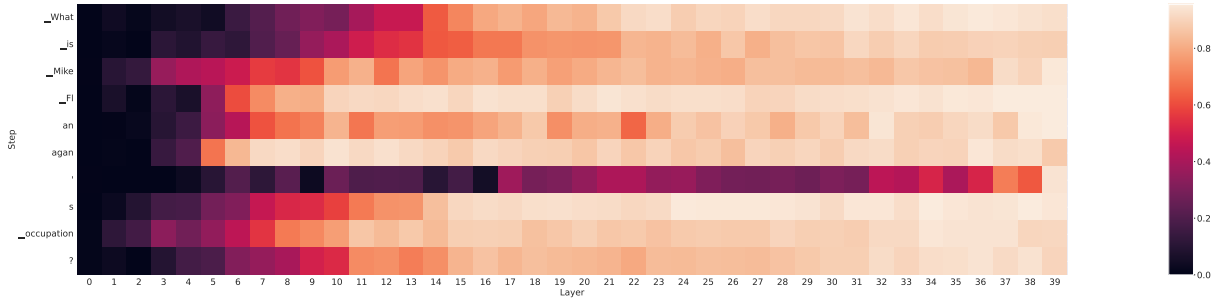
Interestingly, the LLMs with fewer parameters (7B and 13B) reach high $\mathcal{V}$-information earlier, possibly as a result of their forced behavior to generate the correct final output. Therefore, the last few layers of these LLMs encode crucial information for the task. Conversely, the $\mathcal{V}$-information of LLaMA 70B exhibits a slow rate of increase and even demonstrates a decrease in the last few layers. On one hand, LLaMA 70B benefits from a sufficient number of layers to act as a buffer, eliminating the necessity to encode a substantial amount of context knowledge in the last few layers, thereby enhancing its robustness. On the other hand, the last few layers of LLaMA 70B may be employed to eliminate shortcuts in order to enhance contextual comprehension, leading to a marginal reduction in $\mathcal{V}$-information.

Factual	Mike Flanagan is widely recognized and celebrated for his impressive career as a film director, screenwriter, producer, and editor. With numerous accolades and critical acclaim, Flanagan has established himself as a prominent figure in the industry. **...** With his multifaceted skills and successful filmography, it is clear that Mike Flanagan is indeed a talented film director, screenwriter, producer, and editor.
Counterfactual	Mike Flanagan, a highly skilled and talented individual, has made outstanding contributions to the field of graphic design throughout his career. As a graphic designer, Mike has demonstrated exceptional proficiency in various design software, such as Adobe Illustrator and Photoshop. **...** It is undeniable that his vast experience and exceptional skills make him an invaluable asset in the field.

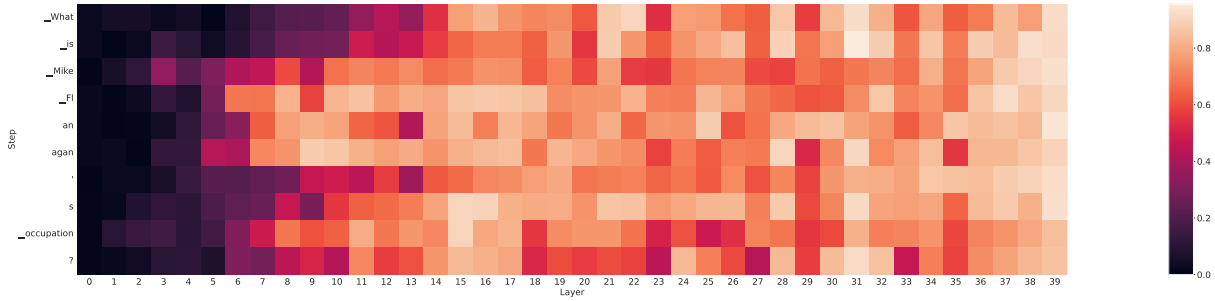
Table 2: Examples of the factual and counterfactual evidence for the case study *What is Mike Flanagan's occupation*. We omit the middle part of the content.



(a) Transformer Layer Output



(b) MLP Layer Output



(c) Attention Layer Output

Figure 3: The heatmap of probing results on LLaMA 2 Chat 13B. We select the question *What is Mike Flanagan's occupation?* as a case study and display the layer-wise $\mathcal{V}$-information for each token in the question.

5.3.3. Knowledge-Related Entity Tokens vs. Other Tokens

To verify the distinction encoded by the LLM between knowledge-related entity tokens and other tokens, we classify the tokens in all questions into two categories. Knowledge-related entity tokens are composed of the subjects and relations associated with knowledge. For example, in the question *What is London the capital of*, *London* and *capital* are knowledge-related entity tokens (Fig. 5). We sequentially conduct the probing task for each token in the 50 constructed questions, obtaining $\mathcal{V}$-information $I_{\mathcal{V}_e}$ for knowledge-related entity tokens and $I_{\mathcal{V}_n}$ for other tokens. The layer-wise differences in means between $I_{\mathcal{V}_e}$ and $I_{\mathcal{V}_n}$ are depicted in Fig. 6.

As can be seen in the figure, knowledge-related entity tokens exhibit significantly higher values of $\mathcal{V}$-